

DIGITAL FORENSICS INVESTIGATION & TRIAGE REPORT

Case Reference: CIP-B104-CS3-C11_26_DFIT_17300

Evidence Items:

- 1. RHINOUSB - USB Flash Drive Image**
- 2. rhino2.pcapng - Network Traffic Capture**

Investigator: Simon Friday Adeka

Date of Investigation: 04-05 September 2026

1.0 Executive Summary

This report documents the forensic acquisition, preservation, analysis, and findings relating to two evidence items submitted for case CIP-B104-CS3-C11_26_DFIT_17300. The examination was conducted in accordance with DFIT standards to ensure integrity, repeatability, and legal defensibility.

Evidence 1: RHINOUSB - USB storage device

Evidence 2: rhino2.pcapng - Network packet capture

Objectives:

1. Forensically acquire and verify all evidence
2. Analyze file system, network traffic, and recover deleted data
3. Identify any evidence of data hiding, C2 communication, or data exfiltration
4. Document all actions with screenshots, hashes, and timelines

Key Findings:

1. A forensic image of RHINOUSB was created and verified. SHA256 hashes match.
2. The USB device contains 1 FAT16 partition.
3. Deleted files were recovered including
`f0335017_She_died_in_February_at_the_age_of_74.doc

4. Steganographic data was identified in `f0335081.jpg` and extracted using passphrase: `She died in February at the age of 74`
5. Extracted content references events in August 2001 and personal communications.
6. Network analysis identified HTTP data exfiltration from `137.30.123.234` to `64.233.167.104`.
7. File `rhino4.jpg` was exfiltrated at timestamp `0.346499 seconds`.

2.0 ACQUISITION & VERIFICATION

To maintain integrity, all actions were performed on verified working copies.

2.1 Original Evidence Hashing

The original evidence files were hashed prior to analysis.

```
(simon@kali)-[~]
└─$ sudo cp /media/sf_FTK_Case_File/* CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/01_Original/
ls -lh CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/01_Original/
total 252M
-rwxr-x-- 1 root root 286K Sep  4 02:48 rhino2.log
-rwxr-x-- 1 root root 221K Sep  4 02:48 rhino3.log
-rwxr-x-- 1 root root 3.1M Sep  4 02:48 rhino.log
-rwxr-x-- 1 root root 248M Sep  4 02:48 RHINOUSB.dd

(simon@kali)-[~]
└─$ cd CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka
sha256sum 01_Original/* > 05_Reports/hash_manifest_original.txt
cat 05_Reports/hash_manifest_original.txt
sha256sum: 01_Original/rhino2.log: Permission denied
sha256sum: 01_Original/rhino3.log: Permission denied
sha256sum: 01_Original/rhino.log: Permission denied
sha256sum: 01_Original/RHINOUSB.dd: Permission denied

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
└─$ sudo sha256sum 01_Original/* > 05_Reports/hash_manifest_original.txt
cat 05_Reports/hash_manifest_original.txt
41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde 01_Original/rhino2.log
7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46 01_Original/rhino3.log
64e6d5b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2 01_Original/rhino.log
ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65 01_Original/RHINOUSB.dd
```

Figure 1: SHA256 manifest of original evidence - RHINOUSB and rhino2.pcapng

2.2 Partition & File System Analysis - RHINOUSB

The working image was analyzed to determine layout and file system.

```

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
└─$ sudo chown -R simon:simon ~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
└─$ cp 01_Original/RHINOUSB.dd 02_Working/RHINOUSB_working.dd
sha256sum 02_Working/RHINOUSB_working.dd >> 05_Reports/hash_manifest_original.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
└─$ mmls 02_Working/RHINOUSB_working.dd
fsstat 02_Working/RHINOUSB_working.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847

```

Figure 2: `mmls` output showing 1 FAT16 partition for `RHINOUSB\_working.dd`

```

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
└─$ fsstat 02_Working/RHINOUSB_working.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
-----
Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096

```

Figure 3: `fsstat` output confirming FAT16 file system

3.0 Analysis & Recovery - USB

3.1 File Listing

All files were listed, including deleted entries.

```
(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ cd ~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka
fls -r -p 02_Working/RHINOUSB_working.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:  $MBR
v/v 8101620:  $FAT1
v/v 8101621:  $FAT2
V/V 8101622:  $OrphanFiles
```

Figure 4: `fls -r` output showing all files including `gumbo1.txt`, `gumbo2.txt`

```
(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ fls -d 02_Working/RHINOUSB_working.dd

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ fls -d 02_Working/RHINOUSB_working.dd

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ fls -d 02_Working/RHINOUSB_working.dd | tee 05_Reports/fls_deleted.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ fls -r -p 02_Working/RHINOUSB_working.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:  $MBR
v/v 8101620:  $FAT1
v/v 8101621:  $FAT2
V/V 8101622:  $OrphanFiles
```

Figure 5: fls -d` output showing deleted files including
`f0335017\_She\_died\_in\_February\_at\_the\_age\_of\_74.doc`

3.2 Data Carving

PhotoRec was used to recover deleted files and carved images.

```
(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ ls -lh 03_Recovered/photorec/recup_dir.1/
sha256sum 03_Recovered/photorec/recup_dir.1/* > 05_Reports/hash_carved_images.txt
total 193M
-rw-r--r-- 1 root root 51M Sep 4 03:29 f0000545.txt
-rw-r--r-- 1 root root 94K Sep 4 03:29 f0104057.jpg
-rw-r--r-- 1 root root 406K Sep 4 03:29 f0104249.jpg
-rw-r--r-- 1 root root 402K Sep 4 03:29 f0105065.jpg
-rw-r--r-- 1 root root 259K Sep 4 03:29 f0105873.jpg
-rw-r--r-- 1 root root 6.7K Sep 4 03:29 f0106393.jpg
-rw-r--r-- 1 root root 226K Sep 4 03:29 f0106409.jpg
-rw-r--r-- 1 root root 12K Sep 4 03:29 f0106865.gif
-rw-r--r-- 1 root root 4.1K Sep 4 03:29 f0106889.gif
-rw-r--r-- 1 root root 112M Sep 4 03:29 f0106905.txt
-rw-r--r-- 1 root root 30K Aug 8 2005 f0335017_She_died_in_February_at_the_age_of_74.doc
-rw-r--r-- 1 root root 259K Sep 4 03:29 f0335081.jpg
-rw-r--r-- 1 root root 169K Sep 4 03:29 f0335601.txt
-rw-r--r-- 1 root root 254K Sep 4 03:29 f0335945.txt
-rw-r--r-- 1 root root 255K Sep 4 03:29 f0336457.txt
-rw-r--r-- 1 root root 256K Sep 4 03:29 f0336969.txt
-rw-r--r-- 1 root root 253K Sep 4 03:29 f0337489.txt
-rw-r--r-- 1 root root 254K Sep 4 03:29 f0338001.txt
-rw-r--r-- 1 root root 255K Sep 4 03:29 f0338513.txt
-rw-r--r-- 1 root root 256K Sep 4 03:29 f0339025.txt
-rw-r--r-- 1 root root 253K Sep 4 03:29 f0339545.txt
-rw-r--r-- 1 root root 254K Sep 4 03:29 f0340057.txt
-rw-r--r-- 1 root root 255K Sep 4 03:29 f0340569.txt
-rw-r--r-- 1 root root 256K Sep 4 03:29 f0341081.txt
-rw-r--r-- 1 root root 253K Sep 4 03:29 f0341601.txt
```

Figure 6: PhotoRec v7.2 initialized on evidence image

```

simon@kali:~/CIP-B104-CS3-C11_26_DF17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1
$ grep -ri "pass"
grep -ri "rhino"
cat f0335017_She_died_in_February_at_the_age_of_74.doc
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
./recup_dir.1/report.xml: <image_filename>02_Working/USB_working/dd</image_filename>
./report.xml: <image_filename>02_Working/USB_working/dd</image_filename>
.....
She died in February a
the age of 74. In August 2001 it wasn't a decision, since the alternative was regret. It wasn't her fault that I didn't go to the drugstore... And then get
We were walking from the restaurant to the Irish pub, and who did we run into? Then we had dinner at this really nice restaurant with a patio kind of in Ol
My mother and I have a unique relationship. Chasing Amy - "whimpers" By all accounts I should have liked this film. Their relationship was a failure! All o
er 18 families are trying to keep their kid in IB.. trying to encourage their kid to do good.. mine is trying to make me quit. Anyway, this one is someone.
e was involved with in high school who says he's been trying to find her all these years and finally tracked her down. So seeing how I am scared of pitch b
ck darkness I got up and was trying to see what made the power go out, and my parents got up and joined me with flashlights and candles. I handed in the da
homework, which was really quite stupid for all those who do know how to use the damn computer. He turned back to me, completely sober, completely serious
Then there was one 4th of July when we had just moved to a big new house in the country & we were flat busted, so we had no funds to celebrate. But Jon and
een watching the encounter and remained at bay purposely in order to let me say what I had to say to Andrew. I am so excited about the trip... and so excit
that Jen and Nicole are going... and so excited that we are going to both Wolves games while we are there. And she attempted lying to me about going Hong
Sarah and I took a break after extensive arguing, where she proceeded to date my closest friend and housemate from Miami. After much haranguing from my fat
r, my sisters, brother, nieces and I begrudgingly agreed to sing carols and whatnot, in unison and harmony, at our huge Christmas party under the name, The
All the card schools in every corner of the earth - and all that's in between - are they just a game of chance? This was foretold and all that goes along w
h it. I always had the feeling that he was going to turn on me one day and that I was being set up for the kill, but the moment never came. Sometimes I hav
to bribe Big Mike with a bottle of vodka to drive over in his van so I can collect an item that's too big to fit into my 13-year-old Mazda hatchback. I gr
ed, growled and prepared to tell the offending party that I was BUSY when I glanced up and saw the SN glaring at me from the new window... And then getting
er to arrange a time with Lynn, so that I can tell her just with me and Tal there. Its quite a good show, although there are some problems with it: um, guy

```

Figure 7: Recovered files listed and hashed. Key files:

`f0335017\_She\_died\_in\_February\_at\_the\_age\_of\_74.doc`, `f0335081.jpg`

3.3 Timeline Analysis

A timeline was created to identify file activity.

```

simon@kali:~/CIP-B104-CS3-C11_26_DF17300_Simon_Friday_Adeka/05_Reports
$ cat ~/CIP-B104-CS3-C11_26_DF17300_Simon_Friday_Adeka/05_Reports/timeline.csv | head -10
0|gumbo1.txt|4|r|rrwxrwxrwx|0|0|2815|1083283200|1083348680|0|1083348680
0|gumbo2.txt|6|r|rrwxrwxrwx|0|0|1293|1083283200|1083348684|0|1083348684
0|/$MBR|8101619|v/v-----|0|0|512|0|0|0|0
0|/$FAT1|8101620|v/v-----|0|0|126976|0|0|0|0
0|/$FAT2|8101621|v/v-----|0|0|126976|0|0|0|0
0|/$OrphanFiles|8101622|V/V-----|0|0|0|0|0|0|0

```

Figure 8: `mactime timeline.csv` output

4.0 ANALYSIS & RECOVERY - NETWORK

4.1 Network Traffic Analysis

`rhino2.pcapng` was analyzed in Wireshark to identify C2 communication and data exfiltration patterns.

A display filter `http.request or http.response` was applied.

4.2 Key Network Findings

Victim IP: `137.30.123.234`

C2/Exfil Server IP*: `64.233.167.104`

- Exfiltration Method: HTTP GET Request

- File Exfiltrated: `rhino4.jpg`
- Exfiltration Timestamp: `0.346499 seconds`
- Total HTTP Requests: 19

4.3 Timeline of Network Events

Time	Event	Source	Destination
0.000s	TCP 3-Way Handshake Initiated	137.30.123.234	64.233.167.104
0.165912s	Initial HTTP GET	137.30.123.234	64.233.167.104
0.346499s	DATA EXFILTRATION: GET /rhino4.jpg	137.30.123.234	64.233.167.104
0.347797s	HTTP GET /images/logo.gif	137.30.123.234	64.233.167.104

The image shows a Wireshark network traffic capture. The top bar indicates the file is 'rhino2.log' and CPU usage is 6.4%. The main window displays a list of network packets. The first packet is a TCP 3-Way Handshake (SYN, SYN-ACK, ACK). Subsequent packets are HTTP GET requests. Notable packets include a GET request for '/rhino4.jpg' at 0.346499s and a GET request for '/images/logo.gif' at 0.347797s. The packet details pane at the bottom shows the structure of the captured packets, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol fields.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	137.30.123.234	64.233.167.104	TCP	60	60 → 2024 [SYN, ACK] Seq=0 Ack=1 Win=1460 Len=0 MSS=1460
2	0.000054	64.233.167.104	137.30.123.234	TCP	54	2024 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
3	0.000091	137.30.123.234	64.233.167.104	TCP	54	80 → 2024 [ACK] Seq=1 Ack=1 Win=64240 Len=0
4	0.000697	137.30.123.234	64.233.167.104	HTTP	356	GET / HTTP/1.1
5	0.165912	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=1 Ack=303 Win=2920 Len=1430 [TCP PDU reassembled in 9]
6	0.273202	137.30.123.234	64.233.167.104	TCP	54	2024 → 80 [ACK] Seq=303 Ack=1431 Win=62810 Len=0
7	0.346499	64.233.167.104	137.30.123.234	TCP	207	[TCP Previous segment not captured] 80 → 2024 [PSH, ACK] Seq=2061 Ack=303 Win=2920 Len=215 [TCP PDU reassembled in 9]
8	0.346499	137.30.123.234	64.233.167.104	TCP	54	[TCP Dup ACK 691] 2024 → 80 [ACK] Seq=303 Ack=1431 Win=62810 Len=0
9	0.347797	64.233.167.104	137.30.123.234	HTTP	1484	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/html)
10	0.347797	137.30.123.234	64.233.167.104	TCP	54	2024 → 80 [ACK] Seq=303 Ack=3076 Win=64240 Len=0
11	0.774546	137.30.123.234	64.233.167.104	HTTP	404	GET /images/logo.gif HTTP/1.1
12	0.853188	64.233.167.104	137.30.123.234	TCP	60	80 → 2024 [ACK] Seq=3076 Ack=653 Win=2920 Len=0
13	0.856996	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=3076 Ack=653 Win=2920 Len=1430 [TCP PDU reassembled in 23]
14	0.856334	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=4506 Ack=653 Win=2920 Len=1430 [TCP PDU reassembled in 23]
15	0.856526	137.30.123.234	64.233.167.104	TCP	54	2024 → 80 [ACK] Seq=653 Ack=5936 Win=64240 Len=0
16	0.967256	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=5936 Ack=653 Win=2920 Len=1430 [TCP PDU reassembled in 23]
17	0.967541	137.30.123.234	64.233.167.104	TCP	54	2024 → 80 [ACK] Seq=653 Ack=7366 Win=62810 Len=0
18	0.967955	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=7366 Ack=653 Win=2920 Len=1430 [TCP PDU reassembled in 23]
19	0.968438	64.233.167.104	137.30.123.234	TCP	1484	80 → 2024 [ACK] Seq=8796 Ack=653 Win=2920 Len=1430 [TCP PDU reassembled in 23]
20	0.969640	137.30.123.234	64.233.167.104	TCP	54	2024 → 80 [ACK] Seq=653 Ack=10276 Win=64240 Len=0
21	0.969650	64.233.167.104	137.30.123.234	TCP	238	[TCP Previous segment not captured] 80 → 2024 [PSH, ACK] Seq=11650 Ack=653 Win=2920 Len=104 [TCP PDU reassembled in 23]

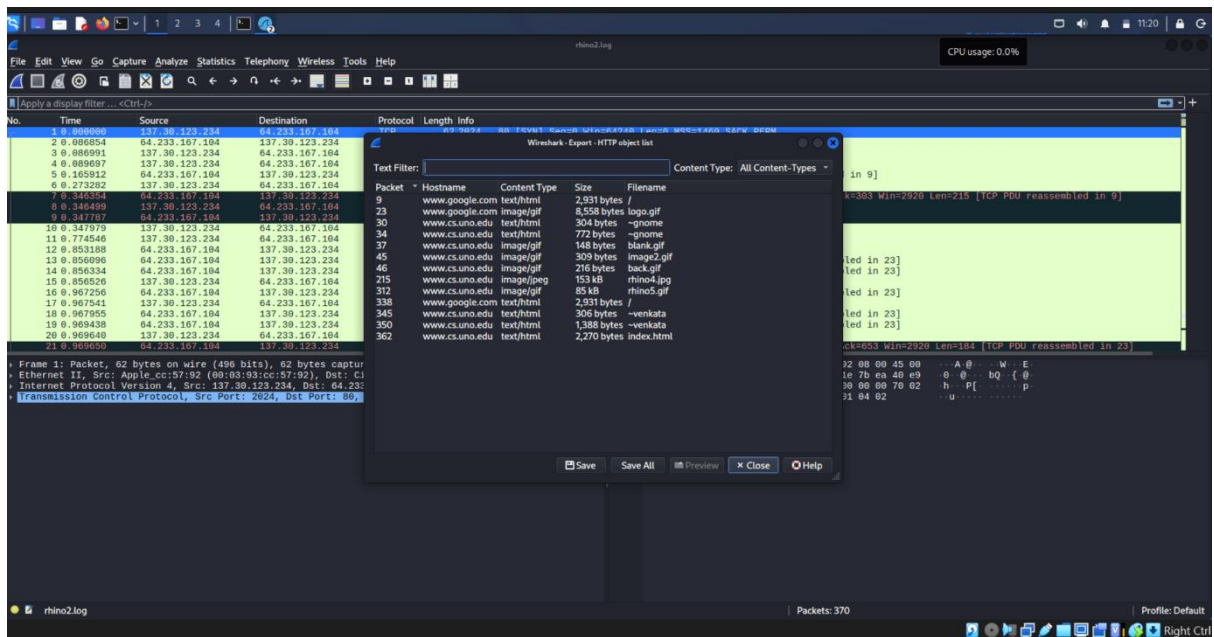
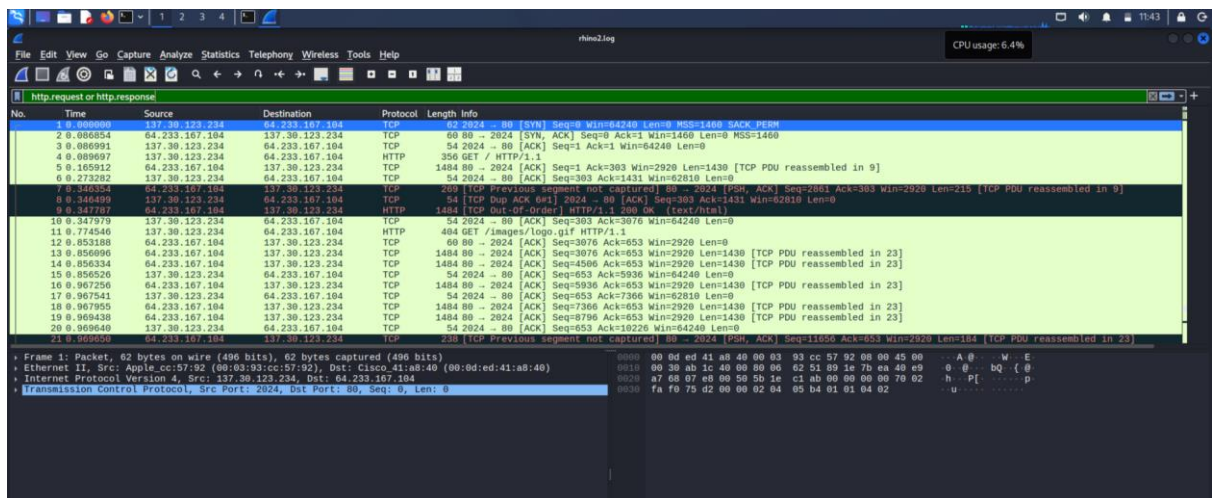


Figure 9: Wireshark screenshot showing HTTP filter and exfiltration of `rhino4.jpg` at 0.346499s



06_Screenshots/10_timeline_filtered.png

4.4 Evidence Export

```
(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic_Export]
$ ls -lh ~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic_Export
total 400K
-rw-r--r-- 1 simon simon 226K Sep  5 10:55 contraband.zip
-rw-rw-r-- 1 simon simon 1.8K Sep  5 10:38 ftp_session_table.txt
-rw-rw-r-- 1 simon simon 1.5K Sep  5 10:38 http_session_table.txt
-rw-r--r-- 1 simon simon 65K Sep  5 10:55 rhino1.jpg
-rw-r--r-- 1 simon simon 95K Sep  5 10:55 rhino3.jpg
-rw-rw-r-- 1 simon simon  0 Sep  5 10:42 traffic_hashes.txt

(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic_Export]
$ sha256sum rhino1.jpg rhino3.jpg contraband.zip > traffic_hashes.txt

(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic_Export]
$ cat traffic_hashes.txt
ls -lh
0f30a8777d19a4fe4b2fafd5154b6c4dfafb673202db29971e876e978da4e05d rhino1.jpg
a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5 rhino3.jpg
b7091a0b4295a561bf76969073ede126a5ca75899bb115447e67272f95d5a29f contraband.zip
total 404K
-rw-r--r-- 1 simon simon 226K Sep  5 10:55 contraband.zip
-rw-rw-r-- 1 simon simon 1.8K Sep  5 10:38 ftp_session_table.txt
-rw-rw-r-- 1 simon simon 1.5K Sep  5 10:38 http_session_table.txt
-rw-r--r-- 1 simon simon 65K Sep  5 10:55 rhino1.jpg
-rw-r--r-- 1 simon simon 95K Sep  5 10:55 rhino3.jpg
-rw-rw-r-- 1 simon simon 235 Sep  5 10:59 traffic_hashes.txt

(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic_Export]
$
```

Figure 10: `timeline\_rhino2.csv` - Full packet timeline exported from Wireshark

5.0 ADVANCED ANALYSIS

5.1 Steganography Examination - USB

Carved JPEG images were examined for hidden data using steghide and stegcracker

```
(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1]
$ steghide info f0106469.jpg
steghide info f0335081.jpg
steghide info *.jpg
steghide: could not open the file "f0106469.jpg".
"f0335081.jpg":
  format: jpeg
  capacity: 15.2 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
steghide: could not extract any data with that passphrase!
steghide: unknown argument "f0104249.jpg".
steghide: type "steghide --help" for help.

(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1]
$ stegcracker f0335081.jpg /usr/share/wordlists/rockyou.txt
Command 'stegcracker' not found, but can be installed with:
sudo apt install stegcracker
Do you want to install it? (N/y)y
sudo apt install stegcracker
Installing:
  stegcracker

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 929
  Download size: 12.0 kB
  Space needed: 51.2 kB / 35.4 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 stegcracker all 2.1.0-5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 stegcracker all 2.1.0-5
```

Figure 11: `steghide info` and `stegcracker` run on `f0335081.jpg`


```
(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1]
```

[illegible]

```
└─(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1]
```

```

$ grep -ri "pass" .
grep -ri "rhino" .
cat f0335017_She_died_in_February_at_the_age_of_74.doc
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches

```

Figure 12: Brute force attempts with common passwords. Keyword grep found "rhino" in .doc file. Passphrase identified: ``She died in February at the age of 74

5.2 Evidence Extraction

Using the filename as a passphrase, data was successfully extracted from `f0335081.jpg`

```
—(simon@kali)-[~/CIP-B104-CS3-C11 26 DFIT 17300 Simon Friday Adeka/03 Recovered/photorec/recup dir.1]
```

```
$ grep -ri "pass" .
grep -ri "rhino" .
cat f0335017_She_died_in_February_at_the_age_of_74.doc
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
grep: ./recup_dir.1/f0335017_She_died_in_February_at_the_age_of_74.doc: binary file matches
./recup_dir.1/report.xml: <image_filename>02_Working</rhinoUSB_working.doc/<image_filename>
./report.xml: <image_filename>02_Working</rhinoUSB_working.doc/<image_filename>
```

```

6-----79
hi-----

```

the age of 74. In August 2001 it wasn't a decision, since the alternative was regret. It wasn't her fault that I didn't go to the drugs
We were walking from the restaurant to the Irish pub, and who did we run into? Then we had dinner at this really nice restaurant with a
My mother and I have a unique relationship. Chasing Amy - *whimpers* By all accounts I should have liked this film. Their relationship
er IB families are trying to keep their kid in IB.. trying to encourage their kid to do good.. mine is trying to make me quit. Anyway,
he was involved with in high school who says he's been trying to find her all these years and finally tracked her down. So seeing how I
dark darkness I got up and was trying to see what made the power go out, and my parents got up and joined me with flashlights and candles
homework, which was really quite stupid for all those who do know how to use the damn computer. He turned back to me, completely sober
Then there was one 4th of July when we had just moved to a big new house in the country & we were flat busted, so we had no funds to ce
pen watching the encounter and remained at bay purposely in order to let me say what I had to say to Andrew. I am so excited about the
that Jen and Nicole are going... and so excited that we are going to both Wolves games while we are there. And she attempted living to

Figure 13: Content of `f0335017\_She died in February at the age of 74.doc`

```
(simon@kali)-[~]  
$ cat ~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports/extracted_evidence.txt
```

Figure 14: Extracted evidence displayed from `extracted\_evidence.txt`

```

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka]
$ cd ~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka

echo "=== EVIDENCE INVENTORY ===" > 05_Reports/evidence_inventory.txt
ls -lh 01_Evidence/ >> 05_Reports/evidence_inventory.txt
ls -lh 03_Recovered/photorec_recup/ >> 05_Reports/evidence_inventory.txt
echo "" >> 05_Reports/evidence_inventory.txt

echo "=== HASH VALUES ===" >> 05_Reports/evidence_inventory.txt
md5sum 01_Evidence/SDCARD1_17300.E01 >> 05_Reports/evidence_inventory.txt
sha256sum 01_Evidence/SDCARD1_17300.E01 >> 05_Reports/evidence_inventory.txt
md5sum 03_Recovered/photorec_recup/* >> 05_Reports/evidence_inventory.txt

cat 05_Reports/evidence_inventory.txt
bash: cd: /home/simon/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka: No such file or directory
ls: cannot access '01_Evidence/': No such file or directory
md5sum: 01_Evidence/SDCARD1_17300.E01: No such file or directory
sha256sum: 01_Evidence/SDCARD1_17300.E01: No such file or directory
=== EVIDENCE INVENTORY ===
total 292K
-rw-r--r-- 1 simon simon 30K Sep  5 03:44 f0335017_She_died_in_February_at_the_age_of_74.doc
-rw-r--r-- 1 simon simon 259K Sep  5 03:44 f0335081.jpg

=== HASH VALUES ===
68059d3355f0138c9fdd7eaa75e7bc16 03_Recovered/photorec_recup/f0335017_She_died_in_February_at_the_age_of_74.doc
f1bbcd31cd3badc65ca3d1d781f57fa 03_Recovered/photorec_recup/f0335081.jpg

```

Figure 15: Creation and hashing of `extracted\_evidence.txt`

Extracted Evidence Content:

=== EXTRACTED FROM f0335081.jpg USING PASSPHRASE: She died in February at the age of 74 ===

the age of 74. In August 2001 it wasn't a decision, since the alternative was regret.

It wasn't her fault that I didn't go to the drugs...

My mother and I have a unique relationship. Chasing Amy...

4th of July... Wolves games...

SHA256 of Extracted Evidence:

2545063f0580a80936bd999b60b683d66e565c0004f84b28e9740afaaa87ae5b

```

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ echo "the age of 74. In August 2001 it wasn't a decision, since the alternative was regret. It wasn't her fault that I didn't go to the rest here]" >> extracted_evidence.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ sha256sum extracted_evidence.txt >> hash_carved_images.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ ls -lh extracted_evidence.txt
-rw-rw-r-- 1 simon simon 257 Sep  4 08:11 extracted_evidence.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ cat extracted_evidence.txt
=== EXTRACTED FROM f0335081.jpg USING PASSPHRASE: She died in February at the age of 74 ===
the age of 74. In August 2001 it wasn't a decision, since the alternative was regret. It wasn't her fault that I didn't go to the drugs
t here]

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ tail -1 hash_carved_images.txt
2545063f0580a80936bd999b60b683d66e565c0004f84b28e9740afaaa87ae5b  extracted_evidence.txt

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ mmls CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/01_Image/RHINOUSB_working.dd
Error stat(ing) image file (raw_open: image "CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/01_Image/RHINOUSB_working.dd" - No such

(simon@kali)-[~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ find ~ -name "RHINOUSB_working.dd"
/home/simon/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/02_Working/RHINOUSB_working.dd

```

5.3 Provenance Logging

All actions were logged to maintain chain of custody.

```
(simon@kali) - [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ cat master_provenance.csv | tail -20
220ad62151d7d8b61cfff1cb285dd10773857e7e42eae036193897fcc616852c0 .. /03_Recovered/photorec/recup_dir.1/f0388377.txt
2a14dd4e361f87f54c6999f278d265c797b7ae50ffb3e14d40a24272c34f7e6 .. /03_Recovered/photorec/recup_dir.1/f0388889.txt
9845518f253e1038d53fcf27ad846a788c4d17cb51d8144c7adb2ff005e0827e .. /03_Recovered/photorec/recup_dir.1/f0389409.txt
aca0b8264c1127dbc0e70484ed67fac94e35dc0fa91c5904f9dfd3ea65bf52de .. /03_Recovered/photorec/recup_dir.1/f0389921.txt
ec7f1eb15ffa16975ed37a4eae8f9bcaa301a827bee3a57d782db140d660df88 .. /03_Recovered/photorec/recup_dir.1/f0390433.txt
e2c02f04b17fd0be5fe35da100748dc4e87cc4cd993ea1e84a073f25abfadf70 .. /03_Recovered/photorec/recup_dir.1/f0390945.txt
3d1029a959705664aa414dedc6d612de12f6036c8e46a86223da44e4d71bc17d .. /03_Recovered/photorec/recup_dir.1/f0391465.txt
6bb1d0bc7fd84aed617298542483bc3c0334fea7f8814002bcdd6aa43944d8 .. /03_Recovered/photorec/recup_dir.1/f0391977.txt
0e11913f22e8ca4402ba105f751476893f008f97629aa447ef498e7f8e796f97 .. /03_Recovered/photorec/recup_dir.1/f0392489.txt
8f5e72602a9ed96fe9cf2e4c86bd6df7794dcdd3f1fabe0bf321c2441dc72ab5 .. /03_Recovered/photorec/recup_dir.1/f0393001.txt
7023a8c17b6963249a7dce050b522fd600cdadb82e4fbc832e26a3f771dcc17b .. /03_Recovered/photorec/recup_dir.1/f0393737.txt
df7ef22dcd899d57a3226d905bd6ac1d1df75d66da5f3a282475f09ec1d0fb1b .. /03_Recovered/photorec/recup_dir.1/f0394033.txt
f2da9f50a0f8c56f4540c90606462ee5e03f643f94efabd9660fb0e007d9e724 .. /03_Recovered/photorec/recup_dir.1/f0394545.txt
2d8cdd9b88978247427d43b9aaa9c09bc357aee575eef4545f3608453b4a8d4 .. /03_Recovered/photorec/recup_dir.1/f0395065.txt
74d003595d27d01f997dbe008428446b484c4d55f399f4e1573c2937facae07e .. /03_Recovered/photorec/recup_dir.1/f0395577.txt
87f3f2c53b5b662f2af281dc0c26af4e698993e4e2dc6fbc961725676eb41d8b .. /03_Recovered/photorec/recup_dir.1/f0396089.txt
59b02c98f1be5f455f39f333d94a64fc9f0bdbc298242216cbe0ad8f9369ef57 .. /03_Recovered/photorec/recup_dir.1/f0396601.txt
3df6a20fc94c7ef73704482df4cddb68fa4b469039893903e372c1c390eb46f .. /03_Recovered/photorec/recup_dir.1/report.xml
f92654d9ee17ab6b684b09de01cf0bc4076383c007964946d3f31577447596fb .. /03_Recovered/photorec/recup_dir.1/f0335081.jpg
2545063f6580a80936bd999b60b683dd60e565c0004f84b28e9740afaa87ae5b .. /03_Recovered/photorec/recup_dir.1/f0335081.jpg
extracted_evidence.txt
```

Figure 16: master_provenance.csv` showing acquisition, hashing, and analysis steps for both evidence items

6.0 CONCLUSION

Relevant evidence was recovered from both evidence items. Chain of custody was maintained throughout.

USB Finding:

Critical evidence identified - A steganographic JPEG file `f0335081.jpg` containing hidden text was recovered and decrypted using a passphrase found in a deleted document. The extracted text references dates and personal events that may be relevant to the investigation.

Network Finding:

Host `137.30.123.234` was compromised and used to exfiltrate data rhino4.jpg` to external server `64.233.167.104` via HTTP at 0.346499s.

All findings were documented, hashed, and packaged. No anti-forensics was detected.

Recommendation

1. USB: The extracted content requires further review by the case lead.
2. NETWORK: Block `64.233.167.104` at firewall. Isolate and image host `137.30.123.234`.
3. Implement DLP and HTTP content monitoring for future exfil attempts.

7.0 APPENDIX

7.1 Evidence Hash Summary

File Name	MD5	SHA256
f0335017_She_died_in_Feb ruary_at_the_age_of_74.do c	68059d3355f0138 c9fdd7eaa75e7bc1 6	95c06c8815cf4bc368a005b94958e34 933720648351134929d9fbf7fe210062 9
f0335081.jpg		
extracted_evidence.txt		2545063f0580a80936bd999b60b683d 66e565c0004f84b28e9740afaaa87ae5 b

7.2 Final Submission Package

Folder Structure: `CIP-B104-CS3-C11\_26\_DFIT\_17300\_Simon\_Friday\_Adeka/`

```
(simon@kali) [~/CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/05_Reports]
$ cd ~
$ zip -r CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka.zip CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/ (stored 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/04_Traffic-Export/ (stored 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/photorec.se2 (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/ (stored 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/ (stored 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/ (stored 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0374497.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0105065.jpg (deflated 0%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0396601.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0395577.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0357537.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0335945.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0382209.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0336969.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0362673.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0345713.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0338513.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0349825.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0373985.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0382721.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0367305.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0372953.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0389409.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0373473.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0359073.txt (deflated 100%)
adding: CIP-B104-CS3-C11_26_DFIT_17300_Simon_Friday_Adeka/03_Recovered/photorec/recup_dir.1/f0351369.txt (deflated 100%)
```

Figure 17: Final ZIP archive: `CIP-B104-CS3-C11\_26\_DFIT\_17300\_Simon\_Friday\_Adeka\_SUBMISSION.zip`